# Ashley Nand

## Data Engineer | Generative AI & LLM Engineer

Strive to excel in designing and deploying Generative AI, LLM, and multi-agent solutions, driving innovation while delivering scalable AI products, cloud-native applications, and end-to-end technology solutions that create measurable business impact and enable transformative user experiences.

+91 9981960193 | Chennai, India | ashnand2000@gmail.com | github.com/PrinceNand | linkedin.com/in/ashley-nand-137286241

Results-driven software engineer with **3+ years of mobile development experience** in **Java** and **Kotlin**, combined with strong expertise in architecting modern Android applications. Skilled in building **AI and GenAI solutions**, including **RAG pipelines**, **multi-agent systems**, and intelligent automation using **Python, LangChain, LlamaIndex**, and leading **LLMs**. Experienced in prompt engineering, retrieval optimization, and developing scalable AI workflows.

## Skills

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| **Languages** | Java, Kotlin, Python, C++, C, TypeScript, JavaScript, SQL |
| **Backend Development** | Spring, Spring MVC, Spring Boot, Node.js, Express.js, RESTful APIs |
| **Mobile Development** | Android Development: Java, Kotlin, XML, Jetpack, Room, SQLite, Firebase |
| **Machine Learning & AI** | Machine Learning: Scikit-learn, NumPy, Pandas, PyTorch; Generative AI: Large Language Models (LLMs), Transformer Architecture, Prompt Engineering |
| **AI Frameworks** | LangChain, LlamaIndex, Streamlit |
| **RAG Systems** | Retrieval-Augmented Generation (RAG), Document Q&A Pipelines |
| **AI Agents & Deployment** | PydanticAI, AutoGen, CrewAI, Multi-Agent Systems, Model Context Protocol (MCP), Ollama, Docker, vLLM, AWS EC2, OpenAI API, Gemini API, Groq API |
| **Frontend Development** | React.js, HTML5, CSS3, Bootstrap, Tailwind CSS, XML |
| **Databases** | MySQL, MongoDB, SQLite, Room, Firebase Realtime Database |
| **Tools & DevOps** | Android Studio, IntelliJ, PyCharm, Git, GitHub, Postman, Docker, Firebase, Ubuntu Server, CI/CD, VS Code |

## Technical Experience

### Software Development Engineer (F2)
*Sep 2023 — Present*
*Mindsprint (Olam Groups)* *Chennai, India*

**Generative AI / Agentic AI Engineer** *Apr 2024 — Present*
**Responsibilities:**

- Designed and implemented LLM-powered applications using GPT models, vector databases, and RAG architectures to automate knowledge retrieval and enhance operational efficiency.
- Built and deployed production-grade AI agents for data extraction, task automation, and workflow orchestration using Python, LangChain, FastAPI, and AWS cloud services.
- Built and tested multi-agent workflows using CrewAI and LangChain, along with a Trader Analyst tool for stock market analysis and trend prediction; containerized projects with Docker and deployed on AWS EC2 as part of hands-on learning.
- Improved model performance through prompt engineering, fine-tuning, dataset curation, and continuous monitoring of accuracy, latency, and cost metrics across deployed AI services.

**OFIS Enumerator (Published on Play Store)** *Sep 2023 — Mar 2024*
**Responsibilities:**

- Developed and launched the OFIS Enumerator Android app for **10+ million farmers** using Java, XML, Jetpack, and Retrofit with Spring Boot APIs for secure, real-time data synchronization.
- Built offline-first architecture with Room and SQLite, improving reliability in low-connectivity regions and reducing crash rates by **40%**.
- Enhanced performance and delivery using Firebase notifications, crash analytics, and optimized CI/CD pipelines with Git, Jenkins, and Gradle for production deployment on Google Play Store.

### TRAINEE – Android Development / OFIS Enumerator (Published on Play Store)
*Feb 2023 — Sep 2023 Olam*
*Information Services Private Limited* *Chennai, India*

- Built and maintained Android applications using Java, XML, Room Database, and SQLite, integrating RESTful APIs and improving performance, offline reliability, and overall code quality.
- Led end-to-end development and issue resolution for multiple apps, including publishing the OFIS Enumerator app on the Google Play Store, while managing version control with Git.

## Publications

**Bi-Directional LSTM Classification of Emotions in EEG Signals** **Published: Jan 16, 2023 - IEEE Xplore**

*2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS)*, IEEE Xplore. Achieved 94.12% accuracy

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## Education

**Master of Technology in Artificial Intelligence & Data Science** (CGPA: 8.9) *April 2023*
*Karunya Institute of Technology & Science*

**Bachelor of Technology in Computer Science & Engineering** (CGPA: 7.9) *August 2021*
*Jawaharlal Nehru Technological University, Kakinada*

**Higher Secondary Education (CBSE) – PCM** (CGPA: 7.2) *May 2017*
*Kendriya Vidyalaya, Bacheli*

## Projects

**Kofi: Study Assistant using CrewAI, RAG, Docker, EC2** *Oct 2025 – Nov 2025*
*Technologies: Python, CrewAI, LlamaIndex, ChromaDB, Streamlit, FastAPI, Groq, Docker, AWS EC2*
Built a RAG-based PDF question-answering system using LlamaIndex and ChromaDB for vector search. Developed a FastAPI + CrewAI backend and a Streamlit frontend, leveraging Groq for fast inference. Containerized with Docker and deployed on AWS EC2, optimizing prompts and embeddings for accurate, context-aware results.

**The Bible Stories** *Nov 2024 – Dec 2024*
*Technologies: MySQL, React, Express, Node.js, Tailwind CSS, Ubuntu, GitHub*
Developed a full-stack web application for managing and publishing Bible-related content. Implemented secure user authentication, optimized back-end APIs, and orchestrated CI/CD pipelines for robust deployment and scalability.

**Time Series Data Analysis – The University of Auckland, New Zealand** *Nov 2021 – May 2023*
*Technologies: Machine Learning, Python, Big Data*
Performed time-series analysis using advanced Python libraries under the mentorship of Dr. Andreas W. Kempa-Liehr. Explored modeling, forecasting, and evaluation techniques for real-world time-dependent data.

**Psychological Emotion-Based Classification of EEG Variations (Data Analytics)** *Oct 2021 – Nov 2022*
*Technologies: Neuroscience, EEG Signals, MATLAB, Big Data*
Conducted EEG signal analysis to classify emotional states by identifying optimal threshold values. Identical patterns extracted from the EEG bands for emotion recognition.

**Product Vending Machine with Machine Learning (IoT Project)** *Mar 2022 – May 2022*
*Technologies: Arduino, Sensor Technology, IoT, Machine Learning*
Built a smart vending machine using Arduino and IoT sensors with RFID authentication. ML was used to analyze seasonal sales data and recommend top-selling products.

**Medicare for e-Health Data Management Technologies (Android App)** *Feb 2021 – Jun 2021*
*Technologies: Android Development, Room DB, Firebase, MVVM*
Designed and developed an Android app to manage patient treatment records. Implemented local storage with Room DB and cloud sync with Firebase for real-time access.

## Certifications

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| NPTEL Online Certification | *Big Data Computing* |
| ICIWRM 2022 | *Int. Conf. on Integrated Water Resources Management* |
| Udemy | *The Data Science Course 2022: Complete Data Science Bootcamp* |
| PSG NI Virtual Instrumentation Center | *Workshop on Real-Time Applications using LabVIEW with Arduino and Raspberry Pi* |
| edX | *Building Modern Node.js Applications on AWS* |
| DevOps Course | *CI/CD (Continuous Integration/Delivery) for Beginners* |
| Microsoft Certification | *Azure Fundamentals (AZ-900)* |
| Udemy | *Learn to Build Microservices using Spring Boot 3, Spring Cloud, React, Kafka, RabbitMQ, REST API, Docker & IntelliJ IDEA* |